

EDUCATIONAL CREDENTIALS

Master of Computer Applications	2024-2026
Vellore Institute of Technology, Chennai	CGPA: 8.86/10
Bachelor of Computer Systems and Design	2021-2024
PSG College of Technology, Coimbatore	CGPA: 8.23/10
HSC, Tamil Nadu State Board	2021
VLMHSS, Coimbatore	Percentage: 93.16%

PROJECTS

Noise-Aware Hybrid Quantum Transfer Learning for Breast Cancer Diagnosis

Tech Stack: Python, PyTorch, PennyLane, OpenCV, NumPy, Scikit-learn, VQC, ZNE

- Developed a hybrid quantum-classical model by integrating EfficientNet-B0 with a variational quantum circuit for breast cancer classification on the BreakHis dataset.
- Built preprocessing and augmentation pipelines, achieving 90% accuracy and evaluating robustness against quantum noise (Pauli, phase damping, depolarizing) using Zero-Noise Extrapolation (ZNE).

FedAdapt: Federated Learning with Riemannian Manifold & ATCNet for EEG Motor Imagery Classification

Tech Stack: Python, PyTorch, Flower, MOABB, ATCNet, Scikit-learn, BCI

- Developed a privacy-preserving federated learning framework for motor imagery EEG classification using ATCNet with Riemannian covariance features on the BCI Competition IV 2a dataset.
- Implemented client-server training, federated aggregation, preprocessing, and evaluation pipelines, achieving 78.8% cross-subject accuracy on non-IID EEG data.

AI-Powered Expense Tracker & Financial Planning System

Tech Stack: Django REST Framework, React, SQLite, Python, REST APIs, NLP, Chart.js/Recharts

- Built a full-stack expense management platform with secure token-based authentication, REST APIs, and NLP-based expense parsing from natural language.
- Developed interactive dashboards for expense analytics, budgeting, savings tracking, and AI-powered financial insights with real-time data visualization.

KEY TECHNICAL SKILLS

Advanced,

Python, SQL, Pandas, NumPy, Matplotlib, PyTorch.

Intermediate,

Java, C++, Scikit-learn, Django, React, GitHub, Power BI, Jupyter Notebook, AWS.

Exposure,

PennyLane, Flower, MOABB, PostgreSQL, MongoDB.

Core Competencies,

Machine Learning, Deep Learning, Computer Vision, Artificial Intelligence, Federated Learning, EDA, Natural Language Processing (NLP), Data Visualization, REST APIs.

MENTORSHIP & VOLUNTEERING

- Placement Representative, PSG College of Technology (2021–2024): Coordinated placement activities and served as the liaison between 60+ students and the placement cell.
- Convenor & Core Member, Animal Welfare Club (2022–2024): Led 20+ volunteers to organize awareness campaigns and community outreach initiatives.
- ERM Lead, PSG College of Technology (2023–2024): Managed logistics and coordinated teams for the successful execution of 8 departmental events.

CERTIFICATIONS

- Data Platform Foundations Associate, Oracle
- AI Engineer for Developers Associate, Datacamp